

An Analytical Study of a Dual-Function Oscillating Water Column with Nonlinear Power Take-Off and a Slotted Supporting Wall [Pre-print]

Clint C. M. Reyes, Zhenhua Huang*

Department of Ocean and Resources Engineering, University of Hawaii at Manoa, Honolulu, 96822 HI, USA

ARTICLE INFO

Keywords:

analytical
eigenfunction expansion
oscillating water column
wave energy
slotted breakwater
renewable energy
non-linear power take-off

ABSTRACT

Combining wave energy converters with coastal structures provides an economical means of converting energy in ocean waves into electricity for nearshore facilities. In this study, we develop an analytical model to assess the performance of a slotted breakwater integrated with a rectangular oscillating water column (OWC) with a nonlinear power-takeoff (PTO). Linear potential flow theory and the eigenfunction matching method are used to obtain the solution. The OWC chamber width, which plays a key role in determining the power-extraction efficiency, was first selected by balancing various design parameters. The effects of the slotted-wall porosity, incident-wave amplitude, and chamber-submergence depth on the performance of the dual-functional structure are also examined. The results show that integrating a slotted breakwater with an OWC significantly enhances the OWC's energy-extraction performance while simultaneously reducing wave transmission through the slotted breakwater, compared with each structure operating alone. The findings of this study offer practical guidance for designing wave-powered harbors that incorporate the proposed dual-functional structures.

1. Introduction

The continued use of conventional energy sources, such as fossil fuels (e.g., coal), pose significant risks that may accelerate the effects of climate change. Over the past few decades, there has been a notable shift of research interest to renewable energy sources (Jakob et al., 2020). Given that the ocean occupies a vast majority of the globe and ocean waves being a dense and predictable source of renewable energy (Clément et al., 2002; Jin and Greaves, 2021), the ability to harness the power of waves could make a significant contribution to the landscape of renewable energy.

One of the most studied wave energy devices is the oscillating water column (OWC) (Falcão and Henriques, 2016). An oscillating water column (OWC) is a partially submerged chamber in which the rise and fall of the water surface from incoming waves forces air through a turbine, generating electricity. Its simplistic design allows for easier maintenance since all the moving parts are not submerged in water.

OWCs have been widely studied in the past few decades (Falcão and Henriques, 2016; Heath, 2012). Evans (1976) developed a linear theory that models the OWC as a weightless piston subjected to sinusoidal waves, assuming the air inside the chamber is incompressible and the internal free surface remains spatially uniform—an approximation valid only when the chamber width is very small compared with the wavelength. Evans (1982) later extended the theory by modeling the spatial variation of surface-pressure fluctuations, allowing the analysis to accommodate chambers whose width is no longer negligible relative to the wavelength. The effects of air compressibility in OWCs was studied by Sarmiento and Falcão (1985), and the results show that the effects of air compressibility is not important for small-scale laboratory OWC models.

Most early studies on OWCs were developed under the assumption of a linear turbine, assuming that there exists a linear relationship between the pressure drop and the flow velocity (e.g. a Wells turbine). Although early studies suggested that using a non-linear turbine offered little improvement in maximum efficiency over a linear turbine (Sarmiento and Falcão, 1985), more recent work (Palmer et al., 2024) shows that a quadratic (impulse) turbine can achieve higher efficiencies across a broader range of wave conditions. Furthermore, using an impulse turbine to achieve the same efficiency as an equivalent Wells turbine would allow for a smaller turbine diameter (Scuotto and de O. Falcão,

*Corresponding author

 ccreyes@hawaii.edu (C.C.M. Reyes); zhenhua@hawaii.edu (Z. Huang)

ORCID(s): 0000-0002-7500-1265 (C.C.M. Reyes)

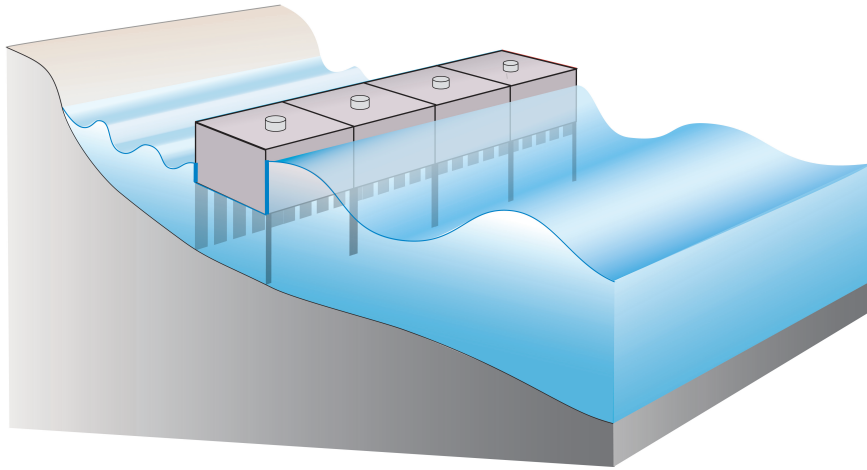


Figure 1: Artistic illustration of the OWC-slotted breakwater (OWC-SB).

2005). Palmer et al. (2024) showed that under strong variations in wave height, a non-linear turbine is capable of maintaining maximal power output; consequently, non-linear turbines offer greater advantages in wave environments that are less stable and more unpredictable. Another advantage of developing models for non-linear turbines is their suitability for representing experimental-scale tests. Because experimental facilities operate at scales much smaller than the prototype, manufacturing a fully functional turbine for small-scale laboratory tests is often impractical. Instead, the power take-off is commonly represented by an orifice, whose behavior is well captured by a non-linear pressure-drop relationship (Mei et al., 1974).

Previous research efforts have led to successful field deployments of OWCs (Torre-Enciso et al., 2009; Suzuki et al., 2004). However, one of the major hurdles to the deployment and commercialization of wave energy converters (WECs) is their high initial cost. One approach to mitigating this challenge is to integrate WECs into existing coastal structures, such as breakwaters, resulting in a dual-purpose design (He et al., 2013). Several studies have demonstrated enhanced OWC performance when the device is installed directly at the coast, which is effectively equivalent to placing the chamber immediately in front of a vertical wall. This configuration has been analyzed analytically using the eigenfunction-expansion matching method (Evans and Porter, 1995a; Martins-rivas and Mei, 2009; Lovas et al., 2010; Zheng et al., 2019; Gang et al., 2022). The vertical wall at the rear enhances power extraction by generating a standing wave directly beneath the OWC. The same conclusion was reached by Xu and Huang (2018); Xu et al. (2025), who examined a three-dimensional circular OWC supported by a C-shaped solid wall, and by Elhanafi et al. (2017); Qu et al. (2023), who compared the performance of OWCs with and without a rear wall. However, adding a vertical solid wall can substantially increase wave loading on the structure, necessitating a thicker wall and potentially deeper foundations, which in turn raises construction costs. Moreover, when the system functions as a breakwater to protect a harbor, the impermeable wall restricts water exchange between the harbor and the open sea, which is undesirable from a water-quality standpoint.

A promising alternative to a solid vertical wall is the slotted or perforated breakwater (Fugazza and Natale, 1992; Bennett et al., 1992; Huang et al., 2011), which uses small gaps to generate turbulent eddies that dissipate wave energy. A related configuration can be created using closely spaced piles, forming what is known as a pile breakwater (Wiegel, 1961; Hayashi et al., 1966). These gaps reduce wave loading on the breakwater and dissipate a significant portion of the incoming wave energy. In addition, this type of breakwater requires less material, potentially lowering construction costs. Furthermore, this design permits water exchange as flow passes through the slots of the breakwater, improving water quality within the harbor and reducing contaminant flushing times. An additional benefit of the slotted breakwater is that the gaps provide passage for marine life. Kakuno and Liu (1993); Bennett et al. (1992); Suh et al. (2006) showed that slotted breakwaters can be analyzed within the framework of potential-flow theory, with structural energy dissipation represented through parameterization and solutions obtained using the eigenfunction-expansion method.

Reyes et al. (2024) recently conducted an experimental study on the integration of an OWC with a slotted breakwater. However, no analytical model currently exists for an OWC combined with a slotted breakwater. Furthermore,

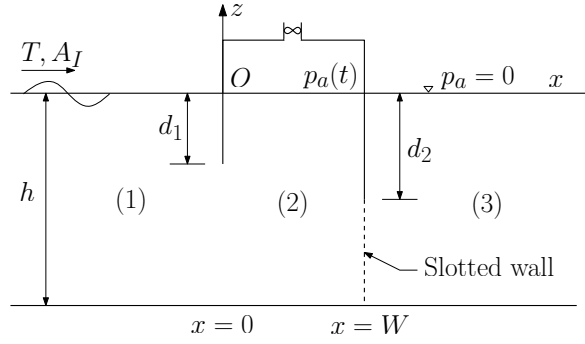


Figure 2: Definition sketch.

as discussed earlier, most previous modeling efforts on OWCs—particularly those integrated into breakwaters—have focused on the linear Wells turbine. Incorporating a nonlinear turbine model would enable the representation of both experimental-scale orifices and more adaptive nonlinear turbines, such as impulse turbines.

In this study, we examine a dual-function slotted breakwater equipped with a rectangular oscillating water column and a nonlinear power take-off system, referred to as the OWC-SB hereafter. An example application of such a dual-purpose structure is illustrated in Fig. 1, which depicts an application for shore protection or a wave-powered harbor incorporating the dual-function integrated system. The analysis is carried out using a theoretical approach, with the solution obtained through potential-flow theory and the eigenfunction-expansion method, as employed by He et al. (2019) and Zheng et al. (2019). The parameterization of the slotted breakwater introduced by Suh et al. (2006) is adopted here to derive the analytical solution. The focus of this study is to examine how the addition of a slotted breakwater and the variation in wave characteristics and structural dimensions influence the pneumatic power-extraction efficiency, wave reflection and transmission, and the energy dissipated by the slotted wall. In addition, we evaluate the wave loading on the structure, an essential consideration for structural design.

2. Problem Formulation

Consider a rectangular OWC integrated with a slotted wall (i.e., an OWC-SB) as shown in Figure 1 and long-crested sinusoidal waves normally approaching the structure. Therefore, the problem is two-dimensional (2D), which can be described in cartesian coordinates with the origin located at the intersection of the still water line with the front face of the structure, the x-axis in the direction of wave propagation and the z-axis pointing upward, as shown in Figure 2. The OWC-SB occupies the region $0 \leq x \leq W$, with the OWC air chamber being bounded by two impermeable walls: the front wall having an immersion depth d_1 at $x = 0$ and the back wall having an immersion depth d_2 at $x = W$. A slotted wall of height $h - d_2$, supports the rear end of the structure at $x = W$.

Wave-interaction with the OWC-SB can be described by potential flow theory with the assumption that flow separation and turbulence generated at the tip of the front face of the chamber can be neglected. Theoretical studies of thin-walled OWCs have shown that sufficiently good results for the determination of OWC designs can be obtained, with some over-prediction in the extraction efficiency of the OWC, due to the neglect of turbulent energy dissipation at the edges of the OWC chamber (Xu et al., 2016).

2.1. Description of wave motion using velocity potential

Consider a train of regular waves with surface displacement expressed as,

$$\zeta_I = \Re \{ A_I e^{-i(\omega t - kx)} \}, \quad \omega = \frac{2\pi}{T}, \quad k = \frac{2\pi}{L}, \quad (1)$$

where, A_I is the incident wave amplitude, T is the wave period, L is the wavelength, t is the time and the operator $\Re \{ \cdot \}$ means taking the real part of its argument. The velocity potential corresponding to the surface displacement of the irrotational incident waves specified by Eq. (1) can be written as

$$\Phi_I = \Re \{ A_I \phi_I e^{-i\omega t} \}, \quad (2)$$

where,

$$\begin{aligned}\phi_I &= \phi_0 e^{ikx} \psi_0(z), \\ \phi_0 &= -\frac{ig}{\omega} \frac{1}{\cosh(kh)}, \quad \psi_0(z) = \cosh(k(z+h)),\end{aligned}\quad (3)$$

h is the water depth, and g is the gravitational acceleration.

Referring to Fig. 2, the problem can be divided into three regions. After parameterizing the local viscous effects of the slotted wall in the potential-flow formulation (see for example Mei et al., 1974; Suh et al., 2006; Huang et al., 2011), the solution can be obtained for the entire flow field. We denote the velocity potential in Region (j) by Φ_j so that the orbital velocity of the wave motion in Region (j) is $\nabla\Phi_j$ and governing equation of motion for each region is simply $\nabla^2\Phi_j = 0$. As will be shown in Sections 2.4 and 2.2, both the slotted wall and the PTO can be linearized so that the problem to be studied is a linear one and the velocity potential in each region can be written as

$$\Phi_j = \Re\{\phi_j e^{-i\omega t}\}, \quad (4)$$

where, ϕ_j is the complex amplitude of Φ_j and a function of x and z for the present problem. Since the structure geometry is simple, ϕ_j can be solved in the frequency domain using an eigen-function matching method with matching conditions provided at the boundaries between two adjacent regions (Linton and McIver, 2001).

2.2. Parametrization of the non-linear PTO

When waves interact with the OWC-SB, the pressure of the air inside the air chamber, denoted by $p_a(t)$, fluctuates. Two kinds of waves exist: waves scattered by the structure due to incoming waves and waves generated by the fluctuating air pressure $p_a(t)$. This fluctuating pressure $p_a(t)$ then drives a turbine, enabling the extraction of wave energy and its conversion into electricity.

In small-scale laboratory experiments, an orifice is commonly used to represent the pneumatic power extraction function of the power take-off (PTO) mechanism. In this case, the air can be safely approximated as an incompressible fluid, and the pressure drop across an orifice can be parameterized in the time domain by

$$p_a(t) = \frac{1}{2} f_{PTO} \rho_a |u(t)| u(t) + \rho_a \ell_{PTO} \frac{du(t)}{dt}, \quad (5)$$

where, $u(t)$ is the cross-sectional average velocity of the air-water interface inside the chamber, f_{PTO} is a coefficient parameterizing the quadratic pressure drop, ℓ_{PTO} a length scale for the oscillatory flow (see for example Xu et al., 2016) and ρ_a is the density of air. To solve the problem in the frequency domain, we linearize the quadratic term in Eq. (5). After introducing the cross-sectional average velocity $u(t) = q(t)/A_{OWC}$ with $q(t)$ being the volumetric flux through the orifice and A_{OWC} the cross-sectional area of the OWC chamber, the linearized version of Eq. (5) can be written as

$$p_a(t) = \rho_a \beta_{PTO} C \frac{q(t)}{A_{OWC}} + \rho_a \frac{\ell_{PTO}}{A_{OWC}} \frac{dq(t)}{dt}, \quad (6)$$

where, C is the phase velocity of the wave and β_{PTO} is a linearized loss coefficient determined by minimizing the mean square error between Eqs. (5) and (6) as described in Xu et al. (2016).

2.3. Frequency-domain representation of the linearized PTO

After linearizing the PTO, the problem is then linear so that the air pressure and the volumetric flux can be expressed as

$$p_a = \Re\{P_a e^{-i\omega t}\}, \quad q(t) = \Re\{Q e^{-i\omega t}\}. \quad (7)$$

Therefore, in frequency domain, Eq. (6) can be rewritten as

$$Q = \frac{A_{OWC}}{\rho_a \beta_{PTO} C - i\omega \ell_{PTO} \rho_a} P_a \equiv (c_{PTO} - i\mu_{PTO}) P_a \quad (8)$$

where

$$c_{PTO} = \frac{A_{OWC} \rho_a \beta_{PTO} C}{(\rho_a \beta_{PTO} C)^2 + (\omega \ell_{PTO} \rho_a)^2},$$

$$\mu_{PTO} = -\frac{A_{OWC} \omega \ell_{PTO} \rho_a}{(\rho_a \beta_{PTO} C)^2 + (\omega \ell_{PTO} \rho_a)^2}$$
(9)

For a linear turbine, c_{PTO} is the conductance, and μ_{PTO} is the susceptance. Note that both c_{PTO} and μ_{PTO} are dimensional. Introducing these parameters into Eq. (8) allows for the validation and verification of our model against the rectangular OWC results of He et al. (2019), which are recovered as a limiting case.

Using Eq. (7), the linearized turbine parameter β_{PTO} can be determined using the principle of equivalent work, i.e.,

$$\beta_{PTO} = \frac{f_{PTO}}{2} \frac{8}{3\pi} \frac{k}{\omega} \frac{|Q|}{A_{OWC}} = f_{PTO} \frac{4}{3\pi} \frac{k}{\omega} \frac{|Q|}{A_{OWC}}$$
(10)

The quadratic loss coefficient f_{PTO} can be estimated using the following formula for a sharp-edged orifice given in Xu et al. (2016) or He and Huang (2017)

$$f_{PTO} = \left(\frac{1}{\sigma C_c} - 1 \right)^2, \quad C_c = 0.6 + 0.4\sigma^2,$$
(11)

where, σ is the opening ratio of the orifice. For incompressible air, the amplitude of the volume flux Q can be expressed in terms of the velocity potential ϕ_2 as

$$Q = B \int_0^W \left. \frac{\partial \phi_2}{\partial z} \right|_{z=0} dx$$
(12)

where, W is the chamber dimension in the along wave direction, B is the chamber dimension along the wave crest. Since the solution of Q is affected by β_{PTO} , an iteration procedure is needed to determine β_{PTO} . The iteration procedure was described in Xu et al. (2016) and is outlined in Section 2.7.

2.4. Frequency-domain representation of the slotted wall

Strictly speaking, the pressure drop across the slotted wall is non-linear. However, Suh et al. (2006) showed that a linear relationship between the pressure drop and the local horizontal velocity can provide satisfactory predictions of the wave reflection and transmission coefficients for a curtain-pile breakwater. In the frequency domain, this linear relationship can be expressed in terms of the velocity potential ϕ_j as

$$\frac{\partial \phi_3}{\partial x} = iG(\phi_2 - \phi_3),$$
(13)

where, ϕ_2 and ϕ_3 are the velocity potentials in Regions (2) and (3) shown in Fig. 2 and G is a parameter describing the approximate linear relationship between the pressure change across the slotted wall and the velocity of the flow at the slotted wall. The expression for G used in Suh et al. (2006) is

$$G = \frac{1}{\frac{\beta_s}{\omega} - i\ell}, \quad \ell = 2C_b.$$
(14)

Let the gap ratio of the slotted wall r_0 be defined by

$$r_0 = \frac{a}{A_s}.$$
(15)

where $2a$ is the width of the gap, $2A_s$ is the center to center distance between two adjacent slots in the slotted wall. The following expressions for β_s and C_b in Eq. (14), given by Kim (1998) and Kakuno and Liu (1993), were adopted by Suh et al. (2002) and Suh et al. (2006):

$$\beta_s = \frac{8f}{9\pi} H_I \omega \frac{1}{\sqrt{(R+2)^2 + P^2}} \frac{5 + \cosh(2kh)}{2kh + \sinh(2kh)},$$
(16)

$$P = \ell k, \quad R = \frac{\beta_s k}{\omega}, \quad H_I = 2A_I, \quad (17)$$

146 and

$$C_b = \frac{b}{2} \left[\frac{1}{r_0} - 1 \right] + \frac{2A_s}{\pi} \left[1 - \log(4r_0) + \frac{1}{3}r_0^2 + \frac{281}{180}r_0^4 \right] \quad (18)$$

147 where b is the wall thickness. The quadratic loss coefficient f for the slotted wall can be determined using the following
148 equation proposed by Mei et al. (1974),

$$f = \left(\frac{1}{r_0 C_c} - 1 \right)^2, \quad C_c = 0.6 + 0.4r_0^2, \quad (19)$$

149 which has been proven applicable to slotted or perforated wave barriers as well (Huang et al., 2011).

150 2.5. Modeling pneumatic power extraction and efficiency using potential flow solutions

151 After linearizing the pressure drop across the orifice, the problem is linear, and the solution can be written as sum
152 of the solution to a radiation problem for waves generated by the pressure fluctuation inside the chamber, p_a in the
153 absence of the incident waves and the solution to a scattering problem for diffracted waves generated by interaction of
154 the incident waves with the structure in the absence of p_a . The total velocity potential is then written as,

$$\phi_j = A_I \hat{\phi}_{S,j} + P_a \hat{\phi}_{R,j}, \quad (20)$$

155 where the subscript S represents the scattering problem, and R represents the radiation problem. Note that by definition
156 the scattering problem is a combination of the incident waves and the waves diffracted by the structure denoted with
157 the subscript D (Newman, 2018):

$$\hat{\phi}_{S,j} = \hat{\phi}_{I,j} + \hat{\phi}_{D,j}. \quad (21)$$

158 The energy produced from the OWC can be obtained from the pressure fluctuations inside the chamber, which is
159 related to the volume flux through Eq. (8).

$$\frac{\partial \phi}{\partial z} = A_I \frac{\partial \hat{\phi}_{S,2}}{\partial z} + P_a \frac{\partial \hat{\phi}_{R,2}}{\partial z}, \quad (22)$$

160 Integrating over the chamber area, the volume flux can be rewritten as

$$Q = A_I \hat{Q}_S + P_a \hat{Q}_R, \quad (23)$$

161 with

$$\hat{Q}_\alpha = B \int_0^W \left. \frac{\partial \hat{\phi}_{\alpha,2}}{\partial z} \right|_{z=0} dx, \quad (24)$$

162 where, $\alpha = S, R$ representing the scattering and radiation fluxes. The radiation flux can be represented with a real and
163 imaginary part (Evans, 1982; Deng et al., 2014; He et al., 2019)

$$Q_R = -(c - i\mu)P_a, \quad (25)$$

164 where, c and μ are real dimensional constants. The normalized radiation flux $\hat{Q}_R$ is then

$$\hat{Q}_R = -(c - i\mu). \quad (26)$$

165 It then follows that the total flux (equation for a Wells turbine) given by Eq. 8 can be written using Eq. 23 as,

$$Q = A_I \hat{Q}_S - P_a(c - i\mu) = (c_{PTO} - i\mu_{PTO})P_a, \quad (27)$$

which gives

$$P_a = \frac{A_I \hat{Q}_S}{(c_{PTO} + c) - i(\mu_{PTO} + \mu)}. \quad (28)$$

For incompressible air, the pneumatic power extraction can be calculated by the work done by the moving air-water interface inside the OWC chamber. The pneumatic power extraction is thus,

$$P_{OWC} = A_{OWC} \overline{u(t)p_a(t)}, \quad (29)$$

where the overbar denotes a time averaging over a wave period. Evaluating Eq. (29) using frequency domain expressions for $u(t)$ and $p(t)$ gives the pneumatic power extracted,

$$P_{OWC} = \frac{c_{PTO}}{2} |P_a|^2. \quad (30)$$

In view of Eq. (28), we have

$$|P_a|^2 = A_I^2 \frac{|\hat{Q}_S|^2}{(c_{PTO} + c)^2 + (\mu_{PTO} + \mu)^2}, \quad (31)$$

and Eq. (30) then becomes,

$$P_{OWC} = \frac{A_I^2}{2} \frac{c_{PTO} |\hat{Q}_S|^2}{(c_{PTO} + c)^2 + (\mu_{PTO} + \mu)^2}. \quad (32)$$

The pneumatic efficiency of the OWC can be defined as

$$\eta = \frac{P_{OWC}}{P_I B}, \quad (33)$$

where, P_I is the incident power over one wavelength per unit wave crest, and B is the OWC dimension along the wave crest. This definition will ensure a dimensionless efficiency of magnitude less than or equal to unity. To determine the pneumatic power and efficiency, the knowledge of β_{PTO} and the solutions of $\hat{\phi}_{S,j}$, $\hat{\phi}_{R,j}$ and P_a are needed.

2.6. Solutions for $\hat{\phi}_{S,j}$ and $\hat{\phi}_{R,j}$ using eigenfunction expansions

The equations governing the velocity potentials $\hat{\phi}_{S,j}$ and $\hat{\phi}_{R,j}$ in Eq. (20) are the following Laplace equations

$$\nabla^2 \hat{\phi}_{\alpha,j} = 0, \quad (34)$$

where $\alpha = S$ for the scattering problem and $\alpha = R$ is for the radiation problem. At the horizontal bed $z = -h$, the no-flux conditions require

$$\frac{\partial \hat{\phi}_{\alpha,j}}{\partial z} = 0, \quad z = -h. \quad (35)$$

On the air-water interface outside the OWC chamber, the combined dynamic and kinematic conditions require

$$-\omega^2 \hat{\phi}_{\alpha,j} + g \frac{\partial \hat{\phi}_{\alpha,j}}{\partial z} = 0, \quad z = 0, \quad j = 1, 3. \quad (36)$$

On the air-water interface inside the OWC chamber, the combined dynamic and kinematic conditions require

$$-\omega^2 \hat{\phi}_{\alpha,2} + g \frac{\partial \hat{\phi}_{\alpha,2}}{\partial z} = \delta_{\alpha,R} \frac{i\omega}{\rho_w}, \quad z = 0, \quad (37)$$

where, $\delta_{R,R} = 1$ and $\delta_{S,R} = 0$ (see Evans and Porter, 1995b; He et al., 2019).

The boundary dividing Regions (1) and (2) is at $x = 0$. The solutions of $\hat{\phi}_{\alpha,1}$ and $\hat{\phi}_{\alpha,2}$ are matched at this boundary. First the no-flux condition is required for the solid plate along the length $-d_1 < z \leq 0$, i.e.,

$$\frac{\partial \hat{\phi}_{\alpha,1}}{\partial x} = \frac{\partial \hat{\phi}_{\alpha,2}}{\partial x} = 0, \quad -d_1 < z \leq 0, \quad (38)$$

where the pressure on either side of the solid plate is unknown. Next, in the region beneath the solid wall, i.e., $-h \leq z < -d_1$, the continuity of velocity and the continuity of pressure require that

$$\frac{\partial \hat{\phi}_{\alpha,1}}{\partial x} = \frac{\partial \hat{\phi}_{\alpha,2}}{\partial x}, \quad -h \leq z < -d_1; \quad (39)$$

$$\hat{\phi}_{\alpha,1} = \hat{\phi}_{\alpha,2}, \quad -h \leq z < -d_1. \quad (40)$$

The boundary dividing Regions (2) and (3) is at $x = W$. The solutions of $\hat{\phi}_{\alpha,2}$ and $\hat{\phi}_{\alpha,3}$ are matched at this boundary. Similarly, the no-flux condition is required along the solid plate $-d_2 < z \leq 0$, i.e.,

$$\frac{\partial \hat{\phi}_{\alpha,2}}{\partial x} = \frac{\partial \hat{\phi}_{\alpha,3}}{\partial x} = 0, \quad -d_2 < z \leq 0, \quad (41)$$

while at the slotted wall, $-h \leq z < -d_2$, the continuity of velocity invokes,

$$\frac{\partial \hat{\phi}_{\alpha,2}}{\partial x} = \frac{\partial \hat{\phi}_{\alpha,3}}{\partial x}, \quad -h < z \leq -d_2. \quad (42)$$

Furthermore, the slotted wall will cause a pressure change, described by Eq. (13), which for scattered and radiated waves can be written, after using the expression for G , as

$$\hat{\phi}_{\alpha,2} - \hat{\phi}_{\alpha,3} = -i \left(\frac{\beta_s}{\omega} - i\ell \right) \frac{\partial \hat{\phi}_{\alpha,3}}{\partial x}, \quad -h < z \leq -d_2. \quad (43)$$

where β_s is given by Eq. (16).

The solutions for the scattering and radiation problems in each region that satisfy the boundary conditions at $z = 0$ and $z = -h$ can be formally written as (see for example Evans and Porter, 1995b; He et al., 2019):

$$\begin{aligned} \hat{\phi}_{\alpha,1} &= \delta_{\alpha,D} \phi_0 e^{ikx} \psi_0(z) + \sum_{m=0}^{\infty} R_{\alpha,m} e^{\lambda_m x} \psi_m(z), \\ \hat{\phi}_{\alpha,2} &= \sum_{m=0}^{\infty} [A_{\alpha,m} e^{-\lambda_m x} + B_{\alpha,m} e^{\lambda_m(x-W)}] \psi_m(z) + \frac{i\delta_{\alpha,R}}{\rho_w \omega}, \\ \hat{\phi}_{\alpha,3} &= \sum_{m=0}^{\infty} T_{\alpha,m} e^{-\lambda_m(x-W)} \psi_m(z), \end{aligned} \quad (44)$$

where, $R_{\alpha,m}$, $A_{\alpha,m}$, $B_{\alpha,m}$, $T_{\alpha,m}$ are unknown coefficients, ϕ_0 is the amplitude or coefficient of the incident potential, and $\lambda_0 = -ik$, and the eigenfunctions are given as

$$\psi_m(z) = \cos(\lambda_m(z+h)), \quad m = 0, 1, \dots, \quad (45)$$

where λ_m is the real ($m = 0$) or complex ($m > 0$) wave number for each eigenmode found by solving the following linear dispersion equation (Mei et al., 2005),

$$\omega^2 = -g\lambda_m \tan(\lambda_m h), \quad m = 0, 1, \dots \quad (46)$$

The coefficients in the formal solutions given by Eq. (44) are to be determined using the matching conditions at $x = 0$ and $x = W$. To solve for unknown coefficients, the infinite series in Eq. (44) are truncated to retain only N evanescent modes. Together with the progressive modes in each region, there will be $4N + 4$ unknown coefficients in the formal solutions given by Eq. (44). The algebraic equations governing the unknown coefficients are obtained in the following two steps:

1. Applying the truncated formal solution as described by Eq. (44) in the matching conditions at $x = 0$ and $x = W$.
2. Multiplying each boundary condition by $\psi_n(z)$ with $n = 0, 1, \dots, N$ and integrating the resulting equations over the entire depth to obtain $4N + 4$ sets of simultaneous linear equations for the unknown coefficients.

We note that β_s is given by Eq. (16) and all the coefficients for $\hat{\phi}_{S,j}$ and $\hat{\phi}_{R,j}$ can be obtained independently of β_{PTO} and P_a . However, the full solution ϕ requires the knowledge of β_{PTO} and P_a , which need to be obtained iteratively using Eq. (10) as outlined in Section 2.7 below.

2.7. Procedure to determine β_{PTO} and P_a

After $\hat{\phi}_{S,j}$ and $\hat{\phi}_{R,j}$ are found, $\hat{Q}_S$ and $\hat{Q}_R$, and thus μ and c can be determined as functions of β_{PTO} to which an iterative procedure is necessary until convergence of β_{PTO} is attained. The iteration procedure to determine β_{PTO} and P_a begins with an initial guess of β_{PTO} :

1. Assume an initial guess for β_{PTO} . If we use $Q^{(0)} = \omega A_I A_{OWC}$ to provide an initial guess of Q , then Eq. (10) gives $\beta_{PTO}^{(0)} = f_{PTO} \frac{4}{3\pi} k A_I$.
2. c_{PTO} and μ_{PTO} are obtained by using $\beta_{PTO} = \beta_{PTO}^{(0)}$ in Eq. (9).
3. The pressure P_a is obtained through Eq. (28) using the values of c_{PTO} and μ_{PTO} obtained in Step 2 and the fluxes obtained from solving the scattering and radiation problems.
4. The total volume flux Q given by Eq. (23) is then obtained by using the pressure P_a obtained in Step 3.
5. A new β_{PTO} given by Eq. (10) is obtained by using the total volume flux Q obtained in Step 4, and the following relative error is calculated,

$$\epsilon_\beta = \left| 1 - \frac{\beta_{PTO}^{(0)}}{\beta_{PTO}} \right|. \quad (47)$$

6. If $\epsilon_\beta < 0.01$, the solution is obtained and the iteration is halted. Otherwise, set $\beta_{PTO}^{(0)} = \beta_{PTO}$ and repeat from Step 2.

Once the iteration converges, we then calculate important parameters that describe the performance of the OWC-SB: (i) the reflection and transmission coefficients, (ii) the power extracted and extraction efficiency and (iii) the wave loading on the structure.

2.8. Reflection, transmission and energy loss coefficients

The reflection (C_R) and transmission (C_T) coefficients of the structure are defined as the ratios of the amplitudes of the reflected and transmitted progressive waves to that of the incident wave. These two coefficients help engineers predict how much wave energy is reflected back into the ocean and how much is transmitted to the other side of a structure, both of which directly affect stability and performance of a breakwater. After the solutions are obtained, the reflection and transmission coefficients can be calculated as,

$$C_R = \left| \frac{A_I R_{S,0} + P_a R_{R,0}}{A_I \phi_0} \right|, \quad (48)$$

$$C_T = \left| \frac{A_I T_{S,0} + P_a T_{R,0}}{A_I \phi_0} \right|.$$

Furthermore, using the conservation of energy, an energy loss coefficient C_L can be defined by the following equation

$$C_L = 1 - C_R^2 - C_T^2 - \eta, \quad (49)$$

which states that the amount of energy dissipated by viscous effects is the difference between the amount of energy in the incident waves and the sum of the amount of wave energy in the reflected and transmitted waves and the amount of the energy extracted by the OWC. Note that in physical experiments, $C_L = C_L^{wall} + C_L^{ends}$, where C_L^{wall} is caused by the energy dissipation due to the slotted wall and C_L^{ends} is caused by the energy dissipation due to the vortex shedding at the lower edges of the OWC chamber walls. If a theory cannot consider the effects of C_L^{ends} , the power extraction efficiency will be over-predicted and such over-prediction must be positively related to C_L^{ends} . In the present model, C_L^{ends} is not included in the formulation; therefore, the extraction efficiency η will be over-predicted, as demonstrated by Xu et al. (2016).

2.9. Horizontal force on the structure

The force acting on a vertical solid or partially slotted wall, located at $x = x_w$ and of unit width along the wave crest, can be calculated by

$$F_H = \int_{-h}^0 \Delta p \, dz, \quad (50)$$

where, Δp is the difference between the dynamic pressure on the up-wave side of the vertical plane and the dynamic pressure on the down-wave-side of the vertical plane. Since the dynamic pressure is related to the velocity potential by the linearized Bernoulli's equation (Mei et al., 2005), Eq. (50) can be rewritten as

$$F_H = -\rho_w \int_{-h}^0 \left(\frac{\partial \Phi_+}{\partial t} - \frac{\partial \Phi_-}{\partial t} \right) dz, \quad (51)$$

where the subscripts + and - denote the up-wave and down-wave sides of the vertical plane, respectively. Let

$$F_H = \Re \{ f_H e^{-i\omega t} \}, \quad (52)$$

where f_H is the complex force amplitude. From Eq. (51), we have

$$f_H = i\omega \rho_w \int_{-h}^0 (\phi_+ - \phi_-) dz. \quad (53)$$

On each side of the vertical plane, the velocity potential includes both the scattered wave potential and radiation potential, i.e.,

$$\phi_{\pm} = A_I \hat{\phi}_{\pm,S} + P_a \hat{\phi}_{\pm,R}, \quad (54)$$

and thus the force amplitude f_H can be calculated by

$$f_H = i\omega \rho_w \int_{-h}^0 [A_I (\hat{\phi}_{+,S} - \hat{\phi}_{-,S}) + P_a (\hat{\phi}_{+,R} - \hat{\phi}_{-,R})] dz \quad (55)$$

This will allow us to calculate the amplitude of the force on the front wall at $x = 0$ and on the back wall at $x = W$. The amplitude of the total force is $f_H(x = 0) + f_H(x = x_w)$. A non-dimensional parameter representing the total horizontal force on the structure can be defined as

$$C_D = \frac{|f_H(x = 0) + f_H(x = x_w)|}{\rho_w g A_I h}. \quad (56)$$

3. Results and Discussion

The validation and verification of the model described in Section 2 are carried out using three limiting cases reported in the literature. Following this verification, an optimal chamber width W is identified, and the effects of slotted wall porosity r_0 , chamber draft d and incident wave amplitude A_I are examined. We will also demonstrate how the OWC can improve the performance of a slotted breakwater and how the slotted wall can in turn, improve the energy extraction by the OWC.

3.1. Model verification and validation

The three limiting cases used to perform the model verification and validation are:

1. Two solid vertical walls separated by a finite distance (Stiassnie et al., 1986), representing an open-top OWC chamber without a turbine. This limiting case is used to validate the wave diffraction by the front and back walls of the OWC chamber
2. A curtain breakwater consisting of a solid upper section with a slotted or perforated lower section (Suh et al., 2006). This limiting case is used to validate the energy dissipation caused by the slotted wall.

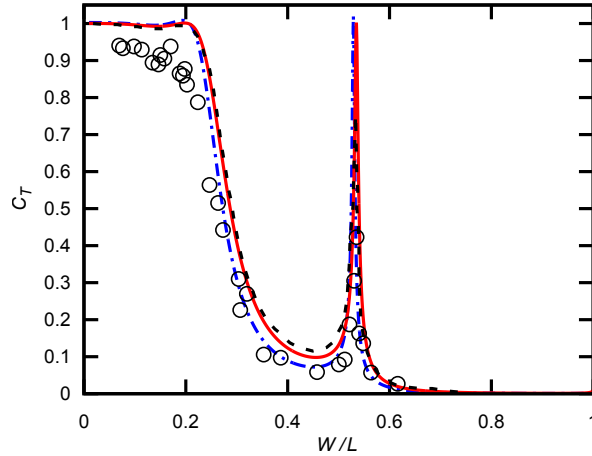


Figure 3: Transmission coefficients obtained from experimental results of Stiassnie et al. (1986) for two surface-piercing thin plates with $W/d = 3$ and $d_1 = d_2 = d$ (circles). The dashed line is for the theoretical results of Stiassnie et al. (1986). The dot-dashed line is for the theoretical results obtained using the present model with $N=20$, and solid line is for the theoretical results obtained using the present model with $N=100$.

3. A rectangular OWC with a linear Wells turbine and variable front and rear wall drafts (He et al., 2019). This limiting case is used to validate the mathematical model of the OWC and the radiated waves generated by the air pressure inside the OWC.

It must be noted that even though both He and Huang (2017) and Reyes et al. (2024) experimentally investigated rectangular OWCs with the PTO represented by orifices, we chose not to use their results for further validation and verification of the present model and code, for the following reasons:

- The present model does not include the effects of the energy loss due to vortex shedding at the lower edges of the OWC chamber walls. As explained in the description of the energy loss coefficient C_L given by Eq. (49), the present model will always over-predict the power extraction efficiency. Therefore, a perfect agreement between the predicted and measured efficiencies does not confirm the correctness of the model or the code; rather, it indicates their incorrectness. As a result, any comparison made will necessarily be subjective and therefore unsuitable for the purpose of verification and validation of the model and code.
- We have applied the present model to the power-extraction-efficiency data reported in He and Huang (2017) and Reyes et al. (2024), and the results show that the model captures the correct trend of the measured power extraction efficiency while consistently over-predicting its magnitude, as expected. Furthermore, the relative difference between the measured and predicted power extraction efficiency is also affected by the model scale. Hence, in order to demonstrate the verification and validation of the model and code, we focus on the theoretical results of He et al. (2019) to demonstrate that the analytical model for the OWC is correct.

3.1.1. Wave diffraction by two surface-piercing thin plates

The first limiting case used to verify the correctness of the model and code is the wave diffraction by two surface-piercing thin plates. The present model can be reduced to this limiting case by setting (i) the opening ratio of the orifice to 1.0 and (ii) the porosity of the rear slotted wall to 1.0.

As shown in Figure 3, the solution converges to the theoretical solution provided by Stiassnie et al. (1986) when the number of the evanescent modes is at least 100. Note that Stiassnie et al. (1986) does not indicate the number of evanescent modes used in their calculations. Nonetheless, using $N = 100$ accurately resolves the physical representation of the transmission coefficient being within 0 and 1. Further tests were performed by increasing N , however minimal changes to the results in C_T are observed for larger values of N . From the results shown in Fig. 3, it is apparent that $N = 100$ is sufficient to represent the wave scattering with two vertical plates.

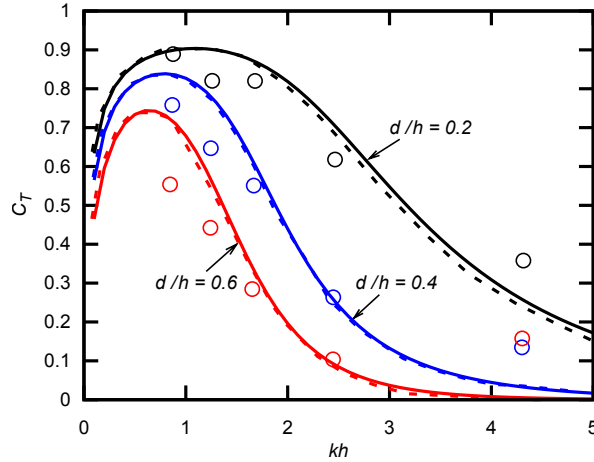


Figure 4: Transmission coefficients obtained from experimental results from Suh et al. (2006) (circles) for a fixed wave steepness of 0.03. The dashed lines are for the theoretical results of Suh et al. (2006) and the solid lines are for the theoretical results obtained using the present model. The three relative curtain drafts are: $d/h = 0.2$, $d/h = 0.4$, $d/h = 0.6$.

On another note, we observe a peak at $W/L \sim 0.5$. This is typically due to resonance with a separation close to half the wavelength, i.e., Bragg resonance of water waves by periodic structures (Mei, 1985; Yu and Chou, 2020). The experimental results of Stiassnie et al. (1986) also show the existence of this peak. Therefore, we can confirm that the resonant peak is observable in nature and is not a result of a numerical singularity. For very long waves, i.e. $W/L \rightarrow 0$, the waves completely pass the structure, while for very short waves $W/L \rightarrow 1$, the structure allows complete reflection.

It is then verified that the present model as formulated in Section 2.6 can correctly solve the diffraction problem without a slotted wall and an OWC air chamber.

3.1.2. A curtain-pile breakwater

The second limiting case corresponds to wave interaction with a curtain-pile breakwater (Suh et al., 2006). The present model can be reduced to this limiting case by setting (i) the draft of the front wall to zero, and (ii) opening ratio of the orifice to 1. The three cases from Suh et al. (2006) are used for comparison in Figure 4 with the following conditions: a water depth of 2.4 m, square piles with $a = 7.15$ cm, $A_s = 14.3$ cm, and $b = 14.3$ cm, corresponding to a slotted-wall porosity of 0.5 according to Eq. (15). Three different drafts of the upper solid section were used: 48 cm, 96 cm, and 144 cm. The number of the evanescent modes used here is 100, as recommended in Section 3.1.1. This choice has shown to be sufficient for comparison with the results of Suh et al. (2006), as illustrated in Fig. 4.

It can be seen that the transmission coefficients obtained using the present theoretical model and measured by Suh et al. (2006) agree well with each other for three values of the relative curtain drafts d/h . It is then verified that the present model can correctly solve the diffraction problem with a slotted wall.

Note that Suh et al. (2006) employed a constant wave steepness of 0.03 in the experiment, which is not realistic in the limit of $kh \rightarrow 0$ and causes the transmission coefficient C_T to decrease in this limit. In contrast, using a constant wave height would make C_T to slowly increase to a constant value less than 1.0.

3.1.3. An rectangular OWC with a Wells turbine in the absence of the slotted wall

The third limiting case is a rectangular OWC with a Wells turbine in the absence of the slotted wall, which was studied analytically by He et al. (2019). The present model can be reduced to this limiting case by setting the porosity of the slotted wall to 1.0. Since a linear Wells turbine exhibits a linear pressure drop, we directly specify c_{PTO} and μ_{PTO} in Eq. (8) as input parameters, instead of treating them as unknown parameters determined by an iteration procedure, as described in Section 2.7.

The results in Figure 4 of He et al. (2019) are used to validate the model implementation of the OWC, the results are shown in Figure 5, which was produced using the following conditions: a water depth of 20 m, front wall draft of $d_1 = 0.15h$, rear wall draft of $d_2 = 0.25h$, chamber width $W = 0.1h$. The PTO parameters corresponding to Figure 4

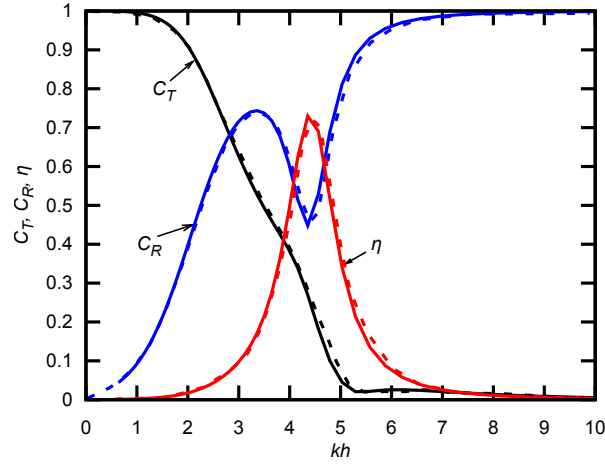


Figure 5: Transmission and reflection coefficients, and pneumatic extraction efficiency for a rectangular OWC with a linear Wells turbine. The dashed lines are the theoretical results of He et al. (2019), and the solid lines are the results obtained by the present model.

of He et al. (2019) are $c_{PTO} = 0.0086 \text{ m}^3 \text{ s} / \text{kg}$ and $\mu_{PTO} = 0.012 \text{ m}^3 \text{ s}^2 / \text{kg} \cdot \omega$. As shown in Figure 5, the coefficients of transmission (C_T) and reflection (C_R) and the pneumatic extraction efficiency (η) produced by the present model agree well with those given in He et al. (2019). It is thus verified that the present model as formulated in Sections 2.5 and 2.6 can solve the radiation problem with a linear PTO.

3.2. Selection of the OWC chamber width

Since wave interaction with the two vertical walls of the OWC chamber generate multiple internal reflections, the chamber width W becomes a critical design parameter that strongly influences the reflection and transmission coefficients, and ultimately the overall energy-extraction efficiency. The optimal OWC chamber width is anticipated to be affected mainly by the design wave period T and water depth h .

Without loss of generality, we will assume that the following design conditions are given: a design water depth of $h = 8.0 \text{ m}$ and a design wave period of $T = 8.0 \text{ s}$. To search for an optimal OWC chamber width for the given design conditions, we examine the ratio W/h over the range of 0.10 to 1.88. In selecting the optimal OWC chamber width, the variability of the wave climate—characterized in this study primarily by changes in the incident wave period—must also be considered. An effective chamber width should allow the device to maintain reliable performance even when the incoming wave period deviates from its design value. We examine results for wave periods ranging from 2.5 s to 55 s, which corresponds to kh ranging from 5 to 0.1 for $h = 8.0 \text{ m}$. Note that the results for very long waves and very short waves are only of academic value.

For the present problem, the slotted wall induces both reflection and dissipation of the incoming wave energy, thereby influencing the overall wave-scattering behavior and the power-extraction performance of the OWC. The optimal width of the OWC chamber must be selected based on a prescribed draft of the OWC chamber and a given porosity of the slotted wall, r_0 , which is the total opening area divided by the total area of the wall. Another name for r_0 is the gap ratio of the slotted wall. In practice, the porosity of the slotted wall decreases over time due to biofouling. Therefore, r_0 should be treated as a time-dependent design parameter. Hard fouling (barnacles, oysters, tubeworms) can grow fast and eventually stabilizes when growth balances mortality and grazing and detachment balance new settlement (Muthukrishnan et al., 2022). We anticipate the value of r_0 will decrease to approximately 0.1 once biofouling stabilizes and we therefore adopt this value when determining the optimal OWC width. The initial gap size of the slots must be designed such that a final porosity of 0.1 can be ensured. The performance of this initial “design structure” will be examined at a later stage of the analysis. Furthermore, bio-fouling will theoretically also influence the draft of the OWC chamber. However, given that the draft is determined by the extreme wave height—which is typically several meters—the change in d caused by bio-fouling is usually relatively small.

In search for the optimal OWC chamber width with $T_p = 8.0 \text{ s}$ and $h = 8.0 \text{ m}$ as the design conditions, we adopt $r_0 = 0.1$ and set $d_1 = d_2 = d = 0.31h$, based on an initial draft $d = 0.3h$ and an assumed biofouling-induced increase

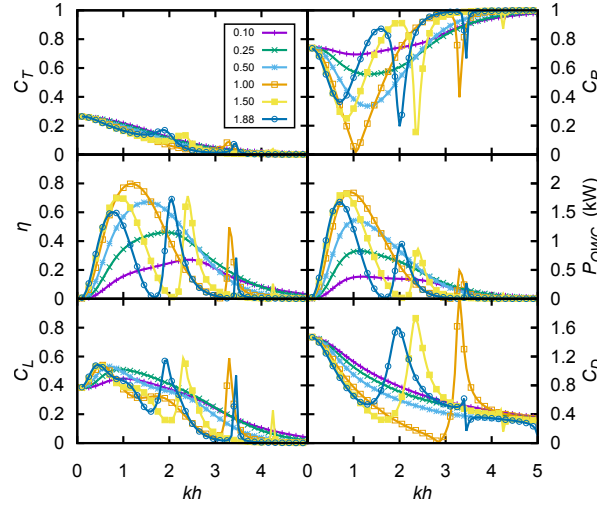


Figure 6: Effects of chamber width, W/h , on the transmission coefficient C_T , reflection coefficient C_R , energy efficiency η , power extracted P_{OWC} , energy dissipation coefficient C_L , and horizontal force coefficient C_D for $r_0 = 0.1$, $A_I/h = 0.11$ and $d/h = 0.31$.

of $0.01h$. For demonstration purposes, we use the following empirical expressions to estimate the significant and root-mean-square wave heights (Mangor et al., 2004),

$$H_s \approx \left(\frac{T_p}{5} \right)^2, \quad H_{rms} = \frac{H_s}{\sqrt{2}}. \quad (57)$$

For $T = 8.0$ s, Eq. (57) gives us $H_{rms} \equiv 2A_I = 1.8$ m, which means $A_I = 0.9$ m, i.e., $A_I/h = 0.11$.

The optimal OWC chamber width will be selected by assessing the following four key performance parameters of the OWC-SB: transmission coefficient C_T , reflection coefficient C_R , the force coefficient C_D , pneumatic efficiency η . Additionally, we examine how the chamber width influences the pneumatic power extracted per meter of the OWC-SB, P_{OWC} , since the chamber width affects not only the extraction efficiency η but also the absolute power generated. Once a fixed chamber width is selected, the trend in the generated power follows that of the extraction efficiency and therefore does not need to be presented in the subsequent analysis. Furthermore, we also examine how the chamber width influences the energy-loss coefficient associated with the slotted wall, C_L .

Figure 6 shows the variation of the four key performance parameters, as well as the energy loss coefficient, C_L and the pneumatic power extracted per meter of the OWC-SB, P_{OWC} , with kh in the range of $kh=0.1-5.0$ for six values of W/h . As expected, the transmission coefficient goes to zero for very large kh for all values of W/h , i.e., the structure will eventually block all the energy of very short waves. The maximum transmission coefficient found at $kh=0.1$ is not affected by the OWC width, and has a value of about 0.26. The peaks and zeros found when kh is greater than 1.8 are due to the resonant interactions of waves with the OWC chamber. Overall, the influence of the OWC width on the transmission coefficient is negligible for practical applications of the designed OWC-SB.

For the reflection coefficient C_R , a W/h of 1.0 produces nearly zero reflection at $kh \sim 1.0$. At $kh=0.77$, which corresponds to the design wave period, both $W/h=1.0$ and $W/h=1.5$ yield similar reflection coefficients of approximately 0.26. All other values of W/h give reflection coefficients larger than 0.26 at the design period. From the perspective of the wave reflection, $W/h=1.0$ is the best OWC width as it can result in small reflection coefficients with a wider range of wave period. The reflection coefficient exhibits more than one local minimum when $W/h \geq 1.0$, an expected outcome due to Bragg resonance (Mei, 1985) or resonant sloshing of the water surface in between the two walls of the OWC chamber (Newman, 1974). However, such local minima may have limited practical significance because the associated wave periods are too short.

The pneumatic power extracted by the OWC, P_{OWC} , exhibits a single peak when $W/h < 1.0$. When $W/h \geq 1.0$, more than one peak exists, with the dominant peak, which has a wider peak width, occurring at a period close to the design wave period. It is clear that the optimal OWC width should be selected from $W/h=1.88$, 1.5 and 1.0, among

which $W/h=1.88$ exhibits the smallest peak magnitude. Even though both $W/h=1.0$ and 1.5 produce comparable peak values, $W/h=1.0$ becomes the preferred option once construction cost is taken into consideration. It is observed that the peak width for $W/h=1.0$ appears wider than that for $W/h=1.5$. However, we note that when looking at the wave period T instead of kh , the two curves produce similar bandwidths close to the peak of the power generation. This is because of the non-linear relationship of T and kh , i.e. as kh decreases from 0.77 to 0.1 , the corresponding wave period increases from 8.0 s to 55 s; however, when kh increases from 0.77 to 5.0 , the wave period decreases only from 8.0 s to 2.5 s.

The variation of the extraction efficiency η with kh is similar to that of the corresponding pneumatic power extraction for each W/h . Although the efficiency for $W/h=1.5$ peaks around the design wave period ($kh = 0.77$), but its peak value is smaller than that for $W/h=1.0$, which peaks around $kh=1.1$. [Ulm et al. \(2023\)](#) commented that, since the scale of the chamber has an effect on P_{OWC} , the design of an OWC should be based on the power extracted rather than on whether its peak efficiency occurs at the design wave period.

With respect to the wave loading, $W/h=1.88$, 1.5 and 1.0 show close to identical C_D values when $kh < 1.0$. However, $W/h=1.0$ is able to maintain a small C_D value under a much wider range of wave periods (up to $kh \sim 3$), and is therefore the preferred option.

The energy dissipation coefficient C_L , which measures the fraction of wave energy lost to turbulence, varies within a narrow range of 0.4 to 0.55 near the design wave period ($kh = 0.77$). The multiple peaks occurring at $kh \geq 2.0$ are due to the resonant interactions between the waves and the OWC-SB. When this type of resonance occurs, more wave energy can be dissipated into turbulence, resulting in a smaller reflection coefficient and a reduced extraction efficiency. However, these resonant conditions occur at very short wave periods, which have limited practical significance.

Overall, we conclude that the optimal chamber width should be $W/h=1.0$ for the given design conditions of wave period $T = 8$ s and water depth $h = 8$ m and for a PTO emulated by an orifice with an opening ratio of 1.25% and a contraction coefficient $C_c = 0.61$. The optimal width of the OWC chamber was determined by balancing wave-power extraction with structural wave loading, local wave-climate variability, construction costs, and the need to control wave reflection and transmission.

Since Bragg resonance ([Mei, 1985](#)) or the resonant sloshing of the water surface between the two chamber walls ([Newman, 1974](#)) is controlled mainly by W/L , the dimensionless analysis given in Appendix A suggests that the results of the four key performance parameters can be presented using the following dimensionless variables for fixed values of W/h , C_f and σ : W/L , A_1/h , d/h and r_0 . In the following sections, we will use the OWC-SB design examined in Fig. 6 with $W/h = 1.0$ as the reference and discuss how variations in slotted-wall porosity, wave amplitude, and OWC chamber draft influence the key performance parameters.

To facilitate the presentation and interpretation of the results in the following sections, we stress that $W/L = 0.12$ corresponds to the design wave period of $T = 8$ s, whereas waves with $W/L \sim 0.5$ are extremely short and therefore hold little practical significance.

3.3. Variation in slotted-wall porosity

As bio-fouling can reduce the slotted-wall porosity, the initial porosity should be designed larger than the desired porosity. The selection of the initial porosity or initial gap size should be based on the expected bio-fouling condition and on an understanding of how the slotted-wall porosity influences the performance of the OWC-SB. For demonstration, we assume an initial porosity of 0.3 , which gradually decreases and stabilizes at 0.1 over time. The changes of the four key performance parameters with the slotted-wall porosity are shown in Figure 7.

As expected, reducing the slotted-wall porosity generally decreases the energy transmitted to the opposite side, while increasing the reflected energy due to the increased blockage effect ([Kakuno and Liu, 1993](#)), which in turn raises the wave force on the structure. Reducing the slotted-wall porosity induces a small shift in the local minimum of C_R to shorter waves or larger W/L . Moreover, as W/L increases, the influence of the porosity gradually diminishes. At the design wave period, reducing the porosity from 0.3 to 0.1 decreases the transmission coefficient from 0.54 to 0.20 , while increasing the reflection coefficient from 0.01 to 0.25 and the force coefficient from 0.42 to 0.94 . Therefore, before bio-fouling stabilizes, the relatively high transmission coefficient must be taken into account during the initial installation period.

With respect to the wave power extraction efficiency, changing the slotted-wall porosity affects only the peak magnitudes, not their locations. Reducing the porosity from 0.3 to 0.1 increases the peak efficiency at $W/L \sim 0.2$ from 0.54 to 0.79 . This increase of the peak efficiency does not seem to be related to the overall reflection coefficient, even though for a circular OWC with a C-shaped support, [Xu et al. \(2016\)](#) found that the extraction efficiency can

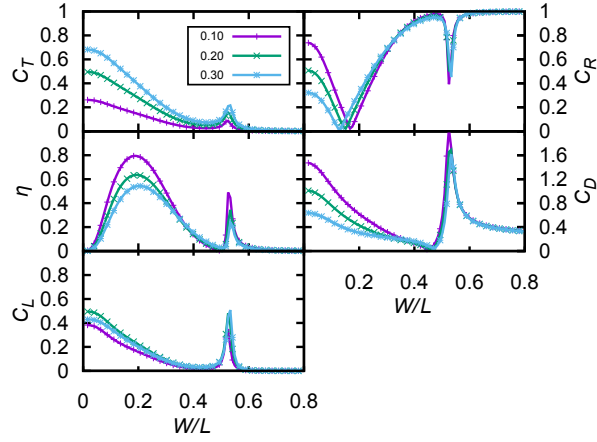


Figure 7: Effects of slotted-wall porosity, r_0 , on the transmission coefficient C_T , reflection coefficient C_R , energy efficiency η and horizontal force coefficient C_D . $W/h = 1.0$, $A_I/h = 0.11$, $d/h = 0.31$.

be increased by the C-shaped support through reflection. Although the porosity can also affect the magnitude of the second peak of the extraction efficiency η at $W/L \sim 0.5$, the influence has less practical significance because the amount of energy that can be extracted is quite small relative to longer waves as shown in Figure 6—such short-period waves typically have small amplitudes and carry less wave power in nature. Therefore, the reduction in slotted-wall porosity caused by bio-fouling is beneficial for wave-energy extraction.

For each porosity, the force coefficient C_D peaks at $W/L \sim 0.5$. On the short wave side of this peak, the force coefficient approaches a constant value. Furthermore, the force coefficient eventually approaches zero at $W/L \sim 0.45$ for all values of porosity. For $W/L < 0.45$, increasing the wave period increases C_D , and eventually at the limit of W/L approaching zero, the force coefficient approach a constant value, which increases under decreasing porosity.

Energy loss to turbulence is measured by the energy loss coefficient C_L , which has a peak at $W/L \sim 0.5$, and gradually decreases toward zero as W/L increases further because when waves are too short—all wave energy is reflected by the front wall and the velocity at the slotted wall is too small to produce any noticeable energy dissipation. It is interesting to note that when $W/L \rightarrow 0$, C_L for each porosity approaches to a constant, which does not change monotonically with the slotted-wall porosity.

The behaviors of C_L and C_D at the limit of $W/L \rightarrow 0$ are related to that fact that the results are obtained with wave amplitude fixed while varying wave period. Physically the energy loss coefficient can also be estimated by the wave power lost to turbulence divided by the incident wave power. The wave power lost to the turbulence can be calculated by the work by the force acting on the slotted wall. For long waves, the horizontal orbital velocity is approximately equal to $(a/h) \cdot C$, where a is the wave amplitude and C is the wave phase velocity. Therefore, the wave force and energy loss will be independent of wave period for given wave amplitude (i.e., long waves are weakly dispersive in the shallow-water limit). This also explains why the force coefficient C_D , or the energy loss coefficient C_L , for each porosity approaches a constant when $W/L \rightarrow 0$.

It can also be observed that resonance occurs at $W/L \sim 0.5$ where C_T , η , C_D and C_L have peaks however C_R has a local minimum. Unlike in He et al. (2019) where the local minimum in the reflection coefficient is due solely to the maximum wave energy extraction at $W/L \sim 0.5$, in our case, due to the presence of the slotted wall, both the wave energy extraction and the energy loss by the slotted wall contribute to the local minimum of C_R . Note also, the values of C_D and C_L at $W/L \sim 0.5$ are larger or equal to their corresponding values in the limit of $W/L \rightarrow 0$. This is because the results are presented with the wave amplitude being fixed while changing the wave period—the wave orbital velocity increases with decreasing wave period once the waves are outside shallow-water wave regime. Therefore, even though the transmission coefficient at $W/L \sim 0.5$ is smaller than its value in the limit of $W/L \rightarrow 0$, the orbital velocity at $W/L \sim 0.5$ may exceed $(a/h) \cdot C$. As a result, the values of C_D and C_L at $W/L \sim 0.5$ are not necessarily smaller than their corresponding values in the limit of $W/L \rightarrow 0$.

To understand further the mechanism by which the slotted-wall porosity influences the power extraction efficiency, we examine how the porosity affects the three parameters in Eq. (32) that characterize the contributions to the power extraction from scattered waves and radiated waves: $|Q_s|$, c and μ . Figure 8 illustrates the influences of slotted-wall

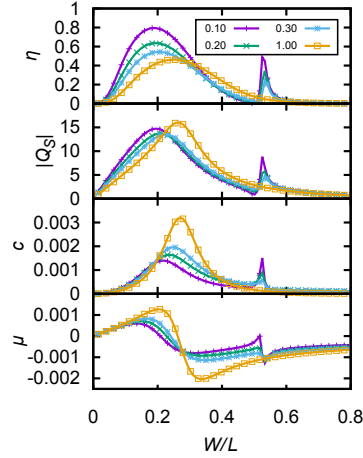


Figure 8: Effects of slotted-wall porosity, r_0 , on energy extraction efficiency η and three key parameters in Eq.(32): $|Q_s|$, c , and μ .

porosity on η , $|Q_s|$, c and μ for four values of porosity: 0.1, 0.2, 0.3 and 1.0. Even though the conductance parameter c is always positive, the susceptance parameter μ changes from positive to negative with increasing W/L . The peak locations of η and Q_s are observed to be very close to each other. The influence of the porosity on c and μ is insignificant when $W/L < 0.15$. Increasing the porosity increases the peak values of c and μ in the range of $0 < W/L < 0.5$.

When Eq. (32) for power extraction is compared with the amplitude response equation for a nonlinearly damped force vibration system (Chakrabarti, 2002), it becomes clear that Q_s plays the role of the external forcing, while the conductance parameter c_{PTO} associated with the PTO and the conductance parameter c associated with the radiated waves act as the effective damping for the OWC system. In our problem, if the PTO is linear, then c_{PTO} is constant; for a nonlinear PTO, c_{PTO} is only weakly affected by the wave conditions. Similar to how the peak of the amplitude response in a nonlinearly damped forced-vibration system is governed primarily by the external excitation and only weakly influenced by damping, the peak of the power-extraction curve is determined mainly by the scattered waves, with the radiated waves exerting only a minor effect.

As for the peak value of extraction efficiency, in the vicinity of the peak of $|Q_s|$, reducing the slotted-wall porosity reduces the values of c and μ , while having only a minor influence on the peak value of $|Q_s|$. This would lead to an increase in the pressure amplitude $|P_a|$ given in Eq. (31), which would then lead to an increase in the power generation and the extraction efficiency.

3.4. Variation in wave amplitude

Since β_s is a function of incident wave amplitude, as shown in Eq. (16), and β_{PTO} is part of the solution and needs to be obtained iteratively (as outlined in Section 2.7), the incident wave amplitude consequently influences the overall performance of the OWC-SB, which makes our theory quasi-linear.

We examine in Figure 9 three different cases of the incoming wave amplitude: 0.5 m, 0.9 m, and 1.8 m, which gives $A_I/h = 0.06$, 0.11, and 0.23, respectively. First of all, we notice that changing the wave amplitude does not alter the positions of the peaks or local minima for any of the four coefficients.

It can be seen that increasing wave amplitude decreases the transmission coefficient. This outcome is expected because Eq. (16) indicates that the resistance of the slotted wall increases as the wave amplitude increases, assuming that all other conditions remain unchanged. This behavior is beneficial as the OWC-SB as a breakwater performs better under larger waves.

For energy extraction efficiency, even though increasing wave amplitude causes a tiny shift in the peak location of η to a smaller W/L or longer waves, the maximum extraction efficiency generally occurs at $W/L \sim 0.2$. The peak value of η increases from 0.65 to 0.80 with increasing A_I/h from 0.06 to 0.23. However, the increase of the efficiency is minimal when A_I/h is raised from 0.11 to 0.23. Note also that increasing the wave amplitude broadens the frequency range over which the efficiency stays greater. This means that more power can be extracted over a wider frequency

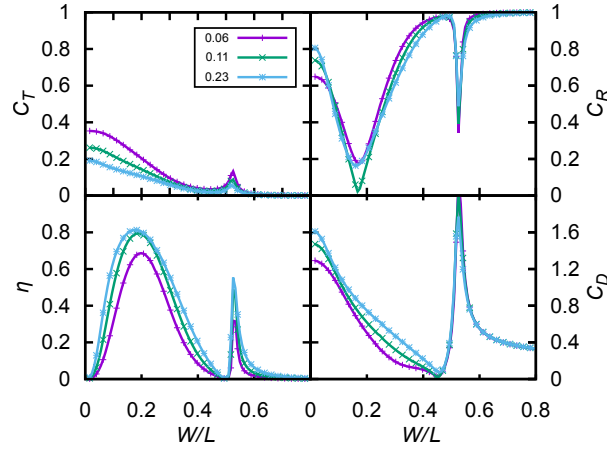


Figure 9: Effects of wave amplitude, A_I/h , on the transmission coefficient C_T , reflection coefficient C_R , energy efficiency η and horizontal force coefficient C_D . $W/h = 1.0$, $r_0 = 0.1$, $d/h = 0.31$.

range under larger waves. Although another peak appears near $W/L \sim 0.5$ for each A_I/h , its magnitude is smaller than the peak at $W/L \sim 0.2$ and the corresponding wave period is too short to have any practical significance.

In the limit of $W/L \rightarrow 0$, the reflection coefficient increases with increasing A_I/h , which is expected as the waves are non-dispersive and the orbital velocity is linearly proportional to A_I/h for the same water depth. Furthermore, Eq. (16) implies that the resistance of the slotted-wall increases with increasing wave amplitude. Among the three different wave amplitudes, the local minimum of the reflection coefficient near $W/L = 0.2$ corresponding to $A_I/h = 0.11$ is the smallest, which is due to the combined effects of wave power extraction and wave transmission.

When $W/L > 0.45$ the wave force coefficient is not significantly affected by the wave amplitude. The force coefficient is close to zero at $W/L = 0.45$, which results from the cancellation of the forces acting on the front and the back members (Rodriguez et al., 2025). When $W/L \leq 0.45$, the force coefficient generally increase slightly with the wave amplitude. It is noted that the influence of the wave amplitude is not significant when $W/L \sim 0.1$, which is close to the design wave period.

3.5. Variation in chamber draft

The results presented in previous sections are for $d/h = 0.31$, which is just an estimated value of a realistic front draft. Theoretically, a shallower draft would lead to better efficiencies (Evans and Porter, 1995a; Li et al., 2022) because a smaller draft allows more wave energy of the incident waves to enter the chamber at a given time. Practically speaking, the lower edges of all chamber walls must remain submerged at all times to prevent air from entering the chamber beneath these edges. Consequently, the required draft should be determined based on the site's tidal variation and extreme wave height, which can be affected by the local bathymetry, wave characteristics and other factors.

Here we examine two other drafts: $d/h = 0.25$ and 0.38 to understand how the chamber draft may affect the four key performance parameters. Fig. 10 shows that varying d/h between 0.25 and 0.38 does not significantly change the shape and trend of the four key performance parameters. The minor influences are summarized below. In general, reducing the draft only slightly increases the transmission coefficient. As expected, increasing the draft leads to slightly higher wave reflection when the incident waves are not in the shallow-water regime (e.g., $W/L < 0.18$). In terms of power extraction efficiency, increasing the draft may reduce performance under conditions where wave dispersion remains significant (e.g., $W/L > 0.18$), which is expected and consistent with the findings of Evans and Porter (1995a) and Li et al. (2022).

Within the range of $0 < W/L < 0.3$, no significant effect of chamber draft on the force coefficient is observed. However, increasing the draft slightly shifts the location of the resonant peak to a lower value of W/L and also increases the peak magnitude. As expected, the force coefficient will not be significantly affected by the chamber draft when $W/L > 0.5$. Note again that waves corresponding to $W/L = 0.5$ are very short and thus have less practical significance.

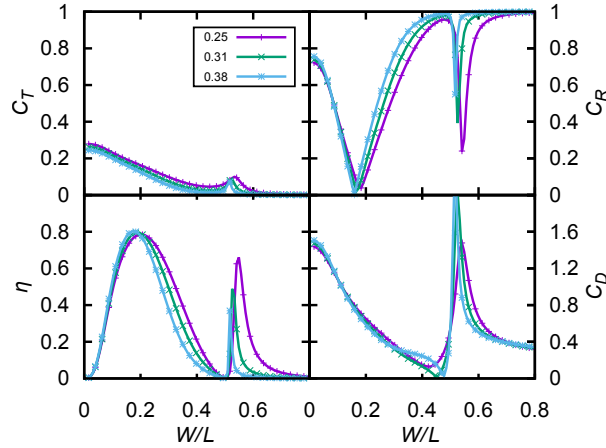


Figure 10: Effects of the chamber draft, d/h on the transmission coefficient C_T , reflection coefficient C_R , energy efficiency η and horizontal force coefficient C_D . $W/h = 1.0$, $r_0 = 0.1$, $A_I/h = 0.11$.

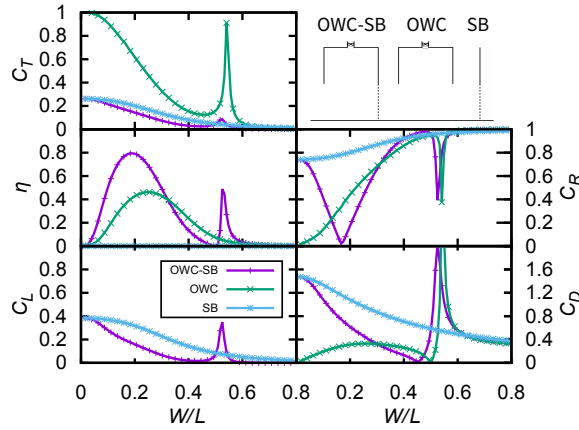


Figure 11: Transmission coefficient C_T , reflection coefficient C_R , energy efficiency η , horizontal force coefficient C_D and slotted wall energy dissipation coefficient C_L for an OWC-SB, a sole OWC and a curtain-pile/slotted breakwater (SB). $A_I/h = 0.11$ and $d/h = 0.31$ for all structures. $W/h = 1.0$ for OWC-SB and the sole OWC. $r_0 = 0.1$ for the OWC-SB and the SB.

Since the four key performance parameters are not very sensitive to variations in chamber draft for the design waves, the initial design and optimization of an OWC-SB breakwater may be focused on the other parameters described in the preceding subsections.

3.6. Performance comparison of the OWC-SB structure with other structures

The reflection and transmission coefficients are key parameters for evaluating the performance of the breakwater, whereas the power extraction efficiency is the primary metric for assessing a wave energy converter. Now we will show that a curtain-pile breakwater or slotted breakwater (SB) (Suh et al., 2006) equipped with an OWC can enhance the breakwater performance and a rectangular OWC (He et al., 2019) using a slotted-wall as its supporting structure can improve its performance for wave energy extraction. Referring to Figure 2, by setting $d_1 = 0$ and removing the PTO, the OWC-SB reduces to a curtain-pile breakwater or a slotted breakwater, while by setting the slotted-wall porosity to 1.0, the OWC-SB reduces to a rectangular OWC. The result of the comparison is summarized in Figure 11.

Comparing an OWC-SB design with a sole OWC, a substantial enhancement in the energy extraction efficiency η is observed, with its peak value rising from 0.45 to 0.8. Moreover, the peak shifts toward a smaller W/L value, being more closer to the design wave condition. The significant increase in the efficiency primarily arises from the slotted wall's ability to substantially enhance the parameter Q_s in Eq. (32) as illustrated in Figure 8. Furthermore, there is a significant drop in the wave transmission coefficient C_T from 0.9 to 0.2 at $W/L = 0.12$, which corresponds to the

design wave period. This is significantly beneficial for harbor or shore protection. However, an increase in the force coefficient C_D is observed for the OWC-SB when $W/L < 0.3$, mainly because of the flow resistance from the slotted wall. Of course, the slotted wall can dissipate a significant amount of wave energy into turbulence as indicated by C_L , which is zero for the sole OWC.

The comparison on breakwater performance of the slotted breakwater (SB) and the OWC-SB will focus only on the wave transmission, reflection and force coefficients. It can be observed that incorporating an OWC significantly reduces both the reflection coefficient and the wave loading within the range $W/L < 0.4$, which is the range of primary practical interest. The wave transmission is also reduced when an OWC is integrated, although the reduction is less pronounced than that observed for wave reflection. For the case of the OWC-SB, the local minimum of the reflection coefficient occurs at the value of W/L for which the extraction efficiency reaches its peak. This implies that the reduction in the wave reflection is strongly affected by wave energy extraction. Another advantage of integrating an OWC with a curtain-pile breakwater is to reduce wave loading associated with the wave reflection, as illustrated by the force coefficient C_D .

Therefore, we can conclude that (1) the addition of an slotted wall to an OWC can improve the OWC's extraction efficiency and (2) the addition of an OWC to a curtain-pile breakwater can reduce the breakwater's transmission and reflection coefficients and lower the wave force on the structure.

4. Conclusions

This study investigates a new OWC design, which is a dual-function wave energy converter integrating a rectangular OWC with a slotted wall. An analytical analysis was performed within the framework of potential flow theory using an eigenfunction expansion approach. The power take-off device was represented by an orifice, whose nonlinear characteristics were linearized for a frequency-domain analysis. The verification and validation of the model against published results for limiting cases confirmed the reliability of the analytical model. The results demonstrate that the chamber width is a key parameter governing energy-extraction performance. The optimal width of the oscillating water column chamber is determined by balancing various design parameters: wave-power extraction with structural wave loading, local wave-climate variability, construction costs, and the need to control wave reflection and transmission. For the problem studied here, it is recommended that the ratio of the OWC chamber width and the local water depth is equal to 1.0. Reducing the porosity of the slotted wall from 0.3 to 0.1 enhances both wave attenuation and OWC power capture, but increases wave reflection and structural loading. Larger incident wave heights were shown to improve power generation and reduce the wave transmission coefficient. The influence of the draft of the OWC chamber on the performance of the dual-function structure is not significant when the ratio of the draft to water depth changes from 0.25 to 0.38. In general, the new dual-function wave energy converter integrating an OWC with a slotted breakwater provides complementary hydrodynamic benefits: the OWC helps to improve the performance of a slotted breakwater and the slotted wall also helps to improve the power extraction efficient of the OWC. The findings of this study provide practical guidance for designing a wave-powered harbor that incorporates the dual-function structure proposed here.

A. Dimensional analysis

To present the results using dimensionless parameters, a simple dimensional analysis is presented here. The transmission and reflection coefficients, force coefficients, and the pneumatic power extraction efficiency depend on the following physical parameters: h , d_1 , d_2 , W , f_{PTO} , f , T_w , A_I , and g . The viscous effects of air and orifice opening ratio σ are implicitly incorporated into $f_{PTO}(Re_a, \sigma)$ with Re_a being a characteristic Reynolds number for the orifice flow and the viscous effects of water and slotted-wall characteristics are implicitly incorporated into $f(Re_w, r_0)$ with Re_w being a characteristic Reynolds number for the flow through the slotted wall. Both f_{PTO} and f are, in principle, functions of Reynolds number but are typically treated as constants, independent of flow conditions, under the assumption that flow separation always occurs (Ulm et al., 2023). In this study, $\sigma = 0.01$ is used so that f_{PTO} is fixed. Since $f(r_0)$ is a monotonic function of r_0 , we will use r_0 to represent f . Using g and h as the repeating variables in the dimensional analysis, we can obtain the following dimensionless relationships,

$$[C_T, C_R, \eta, C_L, C_D] = f\left(\frac{A_I}{h}, \frac{W}{h}, \frac{d_1}{h}, \frac{d_2}{h}, \frac{T_w^2 g}{h}, r_0\right).$$

For a given design, W/h is fixed. We can further replace $T_w^2 g/h$ by $T_w^2 g/W$ using W/h . Using the linear dispersion relationship, we can also use the wave length L to replace the wave period T_w . For the case where $d_1 = d_2 = d$, it then follows that the above dimensionless relationships can be reduced to

$$[C_T, C_R, \eta, C_L, C_D] = f\left(\frac{A_I}{h}, \frac{d}{h}, \frac{W}{L}, r_0\right). \quad (58)$$

The dimensionless width W/L is an important parameter controlling multiple reflections inside the OWC chamber, which may result in Bragg resonance (Mei, 1985). All the results about the effects of wave amplitude, wave period and the chamber draft and width will be presented using the dimensionless parameters identified in Eq. (58).

Acknowledgments

The material presented in this paper is based upon work partially supported by the Naval Facilities Engineering Command (NAVFAC) Engineering and Expeditionary Warfare Center (EXWC), under Contract No. N0002419D6400, Task Order No. N0002421F8805. The views, opinions and/or findings expressed are those of the author and should not be interpreted as representing the official views or policies of the US Navy or the U.S. Government.

References

- Bennett, G., McIver, P., Smallman, J., 1992. A mathematical model of a slotted wavescreen breakwater. *Coastal Engineering* 18, 231–249. URL: <https://www.sciencedirect.com/science/article/pii/037838399290021L>, doi:[https://doi.org/10.1016/0378-3839\(92\)90021-L](https://doi.org/10.1016/0378-3839(92)90021-L).
- Chakrabarti, S.K., 2002. The Theory and Practice of Hydrodynamics and Vibration. volume 20 of *Advanced Series on Ocean Engineering*. World Scientific, River Edge, NJ.
- Clément, A., McCullen, P., Falcão, A., Fiorentino, A., Gardner, F., Hammarlund, K., Lemonis, G., Lewis, T., Nielsen, K., Petroncini, S., Pontes, M.T., Sc, 2002. Wave energy in Europe: current status and perspectives. *Renewable and Sustainable Energy Reviews* 6, 405–431. URL: <https://ideas.repec.org/a/eee/rensus/v6y2002i5p405-431.html>.
- Deng, Z., Huang, Z., Law, A., 2014. Wave power extraction from a bottom-mounted oscillating water column converter with a v-shaped channel. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 470, 20140074–20140074. doi:10.1098/rspa.2014.0074.
- Elhanafi, A., Macfarlane, G., Fleming, A., Leong, Z., 2017. Experimental and numerical measurements of wave forces on a 3d offshore stationary owc wave energy converter. *Ocean Engineering* 144, 98–117. URL: <https://www.sciencedirect.com/science/article/pii/S0029801817305115>, doi:<https://doi.org/10.1016/j.oceaneng.2017.08.040>.
- Evans, D., Porter, R., 1995a. Hydrodynamic characteristics of an oscillating water column device. *Applied Ocean Research* 17, 155–164. URL: <https://www.sciencedirect.com/science/article/pii/0141118795000089>, doi:[https://doi.org/10.1016/0141-1187\(95\)00008-9](https://doi.org/10.1016/0141-1187(95)00008-9).
- Evans, D., Porter, R., 1995b. Hydrodynamic characteristics of an oscillating water column device. *Applied Ocean Research* 17, 155–164.
- Evans, D.V., 1976. A theory for wave-power absorption by oscillating bodies. *Journal of Fluid Mechanics* 77, 1–25. doi:10.1017/S0022112076001109.
- Evans, D.V., 1982. Wave-power absorption by systems of oscillating surface pressure distributions. *Journal of Fluid Mechanics* 114, 481–499. doi:10.1017/S0022112082000263.
- Falcão, A.F., Henriques, J.C., 2016. Oscillating-water-column wave energy converters and air turbines: A review. *Renewable Energy* 85, 1391–1424. URL: <https://www.sciencedirect.com/science/article/pii/S0960148115301828>, doi:<https://doi.org/10.1016/j.renene.2015.07.086>.
- Fugazza, M., Natale, L., 1992. Hydraulic design of perforated breakwaters. *Journal of Waterway, Port, Coastal, and Ocean Engineering* 118, 1–14. doi:10.1061/(ASCE)0733-950X(1992)118:1(1).
- Gang, A., Guo, B., Hu, Z., Hu, R., 2022. Performance analysis of a coast – owc wave energy converter integrated system. *Applied Energy* 311, 118605. URL: <https://www.sciencedirect.com/science/article/pii/S0306261922000824>, doi:<https://doi.org/10.1016/j.apenergy.2022.118605>.
- Hayashi, T., Hattori, M., Kano, T., Shirai, M., 1966. Hydraulic Research on the Closely Spaced Pile Breakwater. pp. 873–884. URL: <https://ascelibrary.org/doi/abs/10.1061/9780872620087.050>, doi:10.1061/9780872620087.050.
- He, F., Huang, Z., 2017. Characteristics of orifices for modeling nonlinear power take-off in wave-flume tests of oscillating water column devices. *Journal of Zhejiang University-SCIENCE A* 18, 329–345.
- He, F., Huang, Z., Law, A.W.K., 2013. An experimental study of a floating breakwater with asymmetric pneumatic chambers for wave energy extraction. *Applied energy* 106, 222–231.
- He, F., Zhang, H., Zhao, J., Zheng, S., Iglesias, G., 2019. Hydrodynamic performance of a pile-supported owc breakwater: An analytical study. *Applied Ocean Research* 88, 326–340.
- Heath, T.V., 2012. A review of oscillating water columns. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 370, 235–245. URL: <https://royalsocietypublishing.org/doi/abs/10.1098/rsta.2011.0164>, doi:10.1098/rsta.2011.0164, arXiv:<https://royalsocietypublishing.org/doi/pdf/10.1098/rsta.2011.0164>.

- Huang, Z., Li, Y., Liu, Y., 2011. Hydraulic performance and wave loadings of perforated/slotted coastal structures: A review. *Ocean Engineering* 38, 1031–1053.
- Jakob, M., Steckel, J., Jotzo, F., Sovacool, B., Cornelsen, L., Chandra, R., Edenhofer, O., Holden, C., Löschel, A., Nace, T., Robins, N., Suedekum, J., Urpelainen, J., 2020. The future of coal in a carbon-constrained climate. *Nature Climate Change* 10. doi:10.1038/s41558-020-0866-1.
- Jin, S., Greaves, D., 2021. Wave energy in the uk: Status review and future perspectives. *Renewable and Sustainable Energy Reviews* 143, 110932. URL: <https://www.sciencedirect.com/science/article/pii/S1364032121002240>, doi: <https://doi.org/10.1016/j.rser.2021.110932>.
- Kakuno, S., Liu, P.L. 1993. Scattering of water waves by vertical cylinders. *Journal of Waterway, Port, Coastal, and Ocean Engineering* 119, 302–322. doi:10.1061/(ASCE)0733-950X(1993)119:3(302).
- Kim, B.H., 1998. Interactions of Waves, Seabed and Structures. Phd dissertation. Seoul National University. Seoul, Korea.
- Li, Y., Liu, S., Xu, C., Li, D., Shi, H., 2022. Experimental study on the cylindrical oscillating water column device. *Ocean Engineering* 246, 110523. URL: <https://www.sciencedirect.com/science/article/pii/S0029801822000038>, doi: <https://doi.org/10.1016/j.oceaneng.2022.110523>.
- Linton, C.M., McIver, P., 2001. Handbook of mathematical techniques for wave/structure interactions. Chapman and Hall/CRC.
- Lovas, S., Mei, C.C., Liu, Y., 2010. Oscillating water column at a coastal corner for wave power extraction. *Applied Ocean Research* 32, 267–283. URL: <https://www.sciencedirect.com/science/article/pii/S0141118710000362>, doi: <https://doi.org/10.1016/j.apor.2010.06.004>.
- Mangor, K., Drønen, N.K., Kærgaard, K.H., Kristensen, S.E., et al., 2004. Shoreline management guidelines.
- Martins-rivas, H., Mei, C., 2009. Wave power extraction from an oscillating water column along a straight coast. *Ocean Engineering - OCEAN ENG* 36, 426–433. doi:10.1016/j.oceaneng.2009.01.009.
- Mei, C.C., 1985. Resonant reflection of surface water waves by periodic sandbars. *Journal of Fluid Mechanics* 152, 315–335. doi:10.1017/S0022112085002154.
- Mei, C.C., Liu, P.L., Ippen, A.T., 1974. Quadratic loss and scattering of long waves. *Journal of the Waterways, Harbors and Coastal Engineering Division* 100, 217–239.
- Mei, C.C., Stiassnie, M., Yue, D.K.P., 2005. Theory and Applications of Ocean Surface Waves. World Scientific. URL: <https://www.worldscientific.com/doi/abs/10.1142/5566>, doi:10.1142/5566, arXiv: <https://www.worldscientific.com/doi/pdf/10.1142/5566>.
- Muthukrishnan, T., Hassenrück, C., Al Fahdi, D., Jose, L., Al Senafi, F., Mahmoud, H., Abed, R.M., 2022. Monthly succession of biofouling communities and corresponding inter-taxa associations in the north-and south-west of the arabian gulf. *Frontiers in Marine Science* 8, 787879.
- Newman, J.N., 1974. Interaction of water waves with two closely spaced vertical obstacles. *Journal of Fluid Mechanics* 66, 97–106. doi:10.1017/S0022112074000085.
- Newman, J.N., 2018. Marine hydrodynamics. MIT press.
- Palmer, H., Zhao, M., Wu, H., Hu, P., Dhamelia, V., 2024. Comparison between linear and quadratic power take off for a single chamber land-fixed oscillating water column (owc). *Renewable Energy* 235, 121329. URL: <https://www.sciencedirect.com/science/article/pii/S0960148124013971>, doi: <https://doi.org/10.1016/j.renene.2024.121329>.
- Qu, M., Yu, D., Li, Y., Gao, Z., 2023. Effect of relative chamber width on energy conversion and mechanical characteristics of the offshore owc device: A numerical study. *Energy* 275, 127372. URL: <https://www.sciencedirect.com/science/article/pii/S0360544223007661>, doi: <https://doi.org/10.1016/j.energy.2023.127372>.
- Reyes, C.C.M., Walker, M., Huang, Z., Cross, P., 2024. A dual-function design of an oscillating water column integrated with a slotted breakwater: A wave flume study. *Energies* 17. URL: <https://www.mdpi.com/1996-1073/17/15/3848>, doi:10.3390/en17153848.
- Rodriguez, A.A.M., Huang, Z., et al., 2025. Wave interaction with two-dimensional submerged perforated thin-plate structures: Effects of structure shape. *Physics of Fluids* 37.
- Sarmiento, A.J.N.A., Falcão, A.F.d.O., 1985. Wave generation by an oscillating surface-pressure and its application in wave-energy extraction. *Journal of Fluid Mechanics* 150, 467–485. doi:10.1017/S0022112085000234.
- Scuotto, M., de O. Falcão, A., 2005. Wells and impulse turbines in an owc wave power plant: a comparison, in: Johnstone, C., Grant, A. (Eds.), Proceedings of 6th European Wave and tidal energy conference, Strathclyde university. pp. 463–469. Neo; 6th Wave and tidal energy conference, Proceedings of an international conference held at the university of Strathclyde, Glasgow ; Conference date: 29-08-2005 Through 02-09-2005.
- Stiassnie, M., Boguslavsky, I., Naheer, E., 1986. Scattering and dissipation of surface wave by a bi-plate structure. *Applied Ocean Research* 8, 33–37. URL: <https://www.sciencedirect.com/science/article/pii/S0141118786800281>, doi: [https://doi.org/10.1016/S0141-1187\(86\)80028-1](https://doi.org/10.1016/S0141-1187(86)80028-1).
- Suh, K.D., Shin, S., Cox, D.T., 2006. Hydrodynamic characteristics of pile-supported vertical wall breakwaters. *Journal of Waterway, Port, Coastal, and Ocean Engineering* 132, 83–96. doi:10.1061/(ASCE)0733-950X(2006)132:2(83).
- Suh, K.D., Son, S.Y., Lee, J.I., Lee, T.H., 2002. Calculation of irregular wave reflection from perforated-wall caisson breakwaters using a regular wave model. pp. 1709–1721.
- Suzuki, M., Arakawa, C., Takahashi, S., 2004. Performance of wave power generating system installed in breakwater at sakata port in japan, pp. ISOPE-I-04–137.
- Torre-Enciso, Y., Ortubia, I., De Aguilera, L.L., Marqués, J., 2009. Mutriku wave power plant: from the thinking out to the reality, in: Proceedings of the 8th European wave and tidal energy conference, Uppsala, Sweden, pp. 319–329.
- Ulm, N., Huang, Z., Cross, P., 2023. Experimental study of a fixed owc-type wave energy converter with a heave plate and v-shaped channels for intermediate-water-depth applications. *Energies* 16. URL: <https://www.mdpi.com/1996-1073/16/16/5988>, doi:10.3390/en16165988.
- Wiegel, R.L., 1961. Closely spaced piles as a breakwater. volume 6. University of California, Hydraulic Engineering Laboratory, Wave Research
- Xu, C., Huang, Z., 2018. A dual-functional wave-power plant for wave-energy extraction and shore protection: A wave-flume study. *Applied Energy* 229, 963–976.

- 718 Xu, C., Huang, Z., Deng, Z., 2016. Experimental and theoretical study of a cylindrical oscillating water column device with a quadratic power take-
719 off model. *Applied Ocean Research* 57, 19–29. URL: <https://www.sciencedirect.com/science/article/pii/S0141118716300256>,
720 doi:<https://doi.org/10.1016/j.apor.2016.02.003>.
- 721 Xu, C., Yang, J., Yao, Y., Zuo, J., Wen, Q., Li, J., 2025. Performance of a closely-spaced array of circular u-owc devices for wave power extraction
722 and breakwater applications. *Ocean Engineering* 324, 120654.
- 723 Yu, X., Chou, H.T., 2020. Bragg resonance of water waves by periodic structures: A review. *Coastal Engineering* 158, 103667. doi:10.1016/j.
724 coastaleng.2020.103667.
- 725 Zheng, S., Zhang, Y., Iglesias, G., 2019. Coast/breakwater-integrated owc: A theoretical model. *Marine Structures* 66, 121–135. doi:<https://doi.org/10.1016/j.marstruc.2019.04.001>.
726